Smart Patient Monitoring and Health Alert System



Problem

- Vital signs are checked manually
- Many readings, concerns are easy to miss
- Goal: flag abnormal SpO_2 for nurses and caregivers

Dataset and motivation

- Kaggle patient monitoring dataset, synthetic
- 60,000 records, 13 columns, no missing values
- Chosen: clean, labelled, beginner-friendly



Algorithm: K-Nearest Neighbors

- KNN compares a patient with 5 similar records
- Predicts the majority class
- Why: simple, explainable, works with numbers



Features and why

Heart rate | **SpO_2** | **Systolic BP** | **Diastolic BP** | **Temperature**

- Target: SpO_2 alert
- SpO_2 alone reproduces the labels
- Scaled so units compare fairly

Prototype demo

[Insert screenshot]

- SpO_2 89%: ABNORMAL
- SpO_2 98%: NORMAL
- SpO_2 110%: rejected

Results

99.225%

test accuracy

	ABNORMAL	NORMAL
ABNORMAL	4,994	52
NORMAL	41	6,913

- 52 missed alerts, 41 false alarms
- Without SpO_2: only 69.53%

Limitations

- Synthetic data, not clinical
- Label is a simple SpO_2 rule
- Missed alerts are risky
- Only SpO_2 range-checked
- Educational use only, not a medical device